

MEDTRACK DV

HOSPITAL PERFORMANCE ANALYTICS DASHBOARD

Dashboard Testing Report

Project Name	MedTrack DV – Hospital Performance Analytics Dashboard
Developer	Madhan Raj G
Internship	Infosys Springboard – Data Visualization Internship
Tool Used	Tableau Public V 2026.2
Dataset	Hospital Operations Dataset (2,238 records)
Document Version	1.0
Report Date	04-Aug-2026

1. Executive Summary

This report documents the dashboard testing activities carried out on the MedTrack DV Hospital Performance Analytics Dashboard prior to final submission under the Infosys Springboard Data Visualization Internship. Testing was conducted across all four published dashboards – Hospital Overview, Patient Flow, Department Analytics, and Resource Utilization – covering functional behavior, KPI accuracy, filter interaction, navigation, and visual consistency.

The testing cycle was completed without any critical or major defects. All executed test cases passed, and every Key Performance Indicator displayed across the dashboard suite was independently recalculated and cross-verified against the source dataset. On the basis of these results, the project is considered stable and ready for final submission.

26	26	0	100%	19
Test Cases Executed	Passed	Failed	Pass Rate	KPIs Validated

2. Project Overview

MedTrack DV is a Hospital Performance Analytics Dashboard developed on Tableau Public as part of the Infosys Springboard Data Visualization Internship. It is built on a hospital operations dataset comprising 2,238 patient records and is designed to give hospital administrators a consolidated, interactive view of operational and clinical performance.

The dashboard suite enables administrators to monitor:

- Patient flow, including admissions, discharges, severity, and length of stay
- Overall hospital performance, including billing, occupancy, and readmission trends
- Department-level analytics, including patient load, revenue, and test outcomes
- Resource utilization, including staffing, bed capacity, and equipment usage
- Core healthcare KPIs used for day-to-day operational decision-making

3. Testing Objectives

The objective of this testing cycle was to independently verify the following aspects of the MedTrack DV dashboard suite before final delivery:

- Verify dashboard functionality across all four dashboards
- Validate KPI calculations against the source dataset
- Validate dashboard navigation between screens
- Validate interactive filters and their downstream effects
- Validate visualizations for accuracy and completeness
- Validate dashboard responsiveness across screen sizes
- Verify underlying business logic behind derived metrics

4. Dashboards Tested

Dashboard	Purpose	Status
Hospital Overview	Consolidated view of patient volume, billing, occupancy, and readmission performance	PASS
Patient Flow	Tracks admissions, severity, discharge status, and patient movement across the hospital	PASS
Department Analytics	Compares patient load, revenue, and clinical outcomes across departments	PASS
Resource Utilization	Monitors staffing, bed capacity, ward allocation, and equipment usage	PASS

5. Testing Summary

Testing Area	Status
Dashboard Loading	PASS
Navigation	PASS
Filters	PASS
KPIs	PASS
Charts	PASS
Tooltips	PASS
Formatting	PASS
Responsiveness	PASS
Performance	PASS

6. KPI Validation Summary

Every KPI listed below was independently recalculated from the source dataset and cross-checked against the value displayed on its respective dashboard.

KPI	Dashboard	Status
Total Patients	Hospital Overview	Validated
Total Billing	Hospital Overview	Validated
Bed Occupancy Rate	Hospital Overview	Validated
Average Length of Stay	Hospital Overview	Validated
Readmission Rate	Hospital Overview	Validated
Discharge Rate	Hospital Overview	Validated
Total Admissions	Patient Flow	Validated
Occupied Beds	Patient Flow	Validated
Severe Cases %	Patient Flow	Validated
ICU Admissions	Patient Flow	Validated
Emergency Cases	Patient Flow	Validated
Average Billing per Patient	Patient Flow	Validated
Total Departments	Department Analytics	Validated
Average Patients per Department	Department Analytics	Validated
Average Billing per Department	Department Analytics	Validated
Total Nurses	Resource Utilization	Validated
Available Beds	Resource Utilization	Validated
Average Patients per Doctor	Resource Utilization	Validated
Total Wards	Resource Utilization	Validated

7. Dashboard Validation Results

7.1 Hospital Overview

All six headline KPIs – Total Patients, Total Billing, Bed Occupancy Rate, Average Length of Stay, Readmission Rate, and Discharge Rate – were verified against the source dataset and displayed accurate values. The Month of Admission, Admission Type, Hospital Region, and Department filters were each tested individually and in combination, with all linked charts and KPI cards refreshing correctly. Navigation to the Home, Dashboards, and About panels functioned as expected, and all nine supporting charts were validated for accuracy and layout.

7.2 Patient Flow

Patient movement charts, including the Monthly Patient Admission Trend and Length of Stay Distribution, were validated against the underlying admission records. Severity analysis was cross-checked against the Severity Distribution chart, and admission analysis, including Age Group, Gender, and Blood Type distributions, was confirmed accurate. The Age Group, Severity, and Discharge Status filters were tested and correctly updated all dependent visuals.

7.3 Department Analytics

Department-level KPIs, including Total Departments, Average Patients per Department, and Average Billing per Department, were recalculated and matched the dashboard exactly. Revenue charts, including Average Revenue by Department and Highest Revenue by Department, were validated against billing totals, and the Medical Condition by Department heatmap was confirmed to reflect accurate patient counts across all six departments.

7.4 Resource Utilization

Staff allocation by shift, bed utilization by ward, and equipment utilization were each verified against the staffing and facilities data. Ward analytics, including Bed Capacity by Ward and ICU Admissions by Ward, were cross-checked against corresponding figures on the Hospital Overview and Patient Flow dashboards and found to be consistent.

8. Defects / Issues

Issue Type	Description	Severity	Status
General QA	No critical issues identified during testing	N/A	Closed
KPI Validation	No KPI mismatches identified	N/A	Closed
Calculations	No broken calculations identified	N/A	Closed
Navigation	No navigation issues identified	N/A	Closed
Filters	No filter issues identified	N/A	Closed
Visualization	No visualization issues identified	N/A	Closed

9. Overall Testing Statistics

Metric	Value
Total Dashboards Tested	4
Total Test Cases	26
Passed	26
Failed	0
Success Rate	100%
KPIs Verified	19
Critical Issues	0

10. Final Conclusion

All four dashboards within the MedTrack DV Hospital Performance Analytics Dashboard have passed testing successfully. Every KPI displayed across the dashboard suite was validated against the source dataset, and no discrepancies were identified. Dashboard navigation and interactive filters function correctly across all modules, and visual consistency has been verified in terms of formatting, layout, and chart accuracy.

Dashboard performance and responsiveness were found to be satisfactory under standard testing conditions. Based on the results documented in this report, the MedTrack DV project is approved for final submission under the Infosys Springboard Data Visualization Internship.